DEV VRAT

devvratalewa@gmail.com | +91 7828943224 | LinkedIn | GitHub

PROFESSIONAL SUMMARY

Computer Science undergraduate (AI/ML focus) with hands-on experience in building and shipping end-to-end software systems spanning computer vision, deep learning, and cloud applications. Proficient in Python, C++, and Java with a strong foundation in data structures, algorithms, and object-oriented design. Proven ability to lead teams and deliver technical projects, backed by AWS and cloud computing certifications and active open-source contribution on GitHub.

EDUCATION

VIT Bhopal University 2023 – 2027 *Bachelor of Technology in Computer Science & Engineering* *Madhya Pradesh, India*

Cumulative GPA: 8.62 / 10.0

TECHNICAL SKILLS

Languages: C, C++, Java, Python

AI / ML: Machine Learning, Deep Learning, CNN, TensorFlow, OpenCV

Frameworks & Libraries: Flask, Node.js, WebRTC

Cloud & Databases: AWS, MongoDB

Developer Tools: Git, GitHub, Linux, Arduino, MATLAB, CI/CD Concepts

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Cloud Computing

PROJECTS

Robotic Prosthetic Hand *Python | OpenCV | Arduino | Embedded Systems*

- Engineered a camera-based, cost-effective prosthetic hand that translates real-time hand gestures into precise servo motor movements for assistive motion control.
- Implemented a real-time computer vision pipeline in OpenCV to detect and classify hand gestures, interfacing model output with Arduino firmware for low-latency actuation.

Deepfake Detection Platform *CNN | Xception | OpenCV | Flask | Node.js | MongoDB | WebRTC*

- Built a full-stack AI platform for real-time image and video authenticity verification, combining a CNN/Xception deep learning backend with a Flask API and Node.js/MongoDB web app.
- Designed an inference pipeline processing live video streams via WebRTC, integrating model serving, storage, and UI into one system for real-time (not post-hoc) deepfake detection.

SecurNet — DDoS Detection & Mitigation *TensorFlow | Deep Neural Networks | Ryu SDN Controller | Mininet | Linux*

- Designed a Deep Neural Network-based intrusion detection system for Software-Defined Networks, achieving 99.98% detection accuracy on simulated DDoS traffic.
- Built and tested the system in a Mininet-emulated topology controlled via the Ryu SDN controller, automating real-time blocking of malicious flows upon detection.

Beam Analysis & Report Generation Tool *Python | LaTeX | Excel Automation*

- Automated structural engineering workflows by building a tool that ingests beam specifications from Excel and computes Shear Force and Bending Moment Diagrams (SFD/BMD) programmatically.
- Generated professional, publication-quality PDF engineering reports using LaTeX, eliminating manual formatting and streamlining the analysis-to-report workflow.

LEADERSHIP & ACTIVITIES

Co-Lead, Corporate Team — Notion Community Club, VIT Bhopal

- Led corporate outreach and sponsorship efforts, coordinating with external partners and a student team to secure collaborations and resources for club initiatives.

Founder — Andy Haryana Club

- Founded and scaled a student community club from the ground up, defining its mission and structure and coordinating its early events and outreach.

Core Team Member — AdVITya'24 (VIT Bhopal Annual Fest) · Member, National Service Scheme (NSS) Unit

- Organized large-scale student events as a core committee member for VIT Bhopal's flagship fest; contributed to community service initiatives through the NSS unit.

CERTIFICATIONS & ACHIEVEMENTS

- **Elite Certification, Cloud Computing** — NPTEL
- **Generative AI using IBM watsonx** — IBM
- **Microsoft Certified: Azure Data Fundamentals (DP-900)** — Microsoft